Project Design Phase
Problem – Solution Fit

Date	16 Feb 2026
Team ID	SWTID-2026-8203
Project Name	MicroGoal
Maximum Marks	2 Marks

Problem – Solution Fit Overview:-

The Problem–Solution Fit ensures that the identified problem matches real user needs and that the proposed backend solution effectively solves those issues before scaling further.

Purpose

- Address the lack of a **secure and structured backend system** for managing personal goals.
- Provide users with a **reliable API** to create, update, delete, and view goals.
- Ensure **secure authentication and authorization** using JWT.
- Improve data protection by restricting access to authorized users only.
- Deliver a scalable backend architecture that supports future feature expansion.

Problem Statement

Many users want to manage their daily or long-term goals digitally, but:

- Existing systems are either not secure or overly complex.
- Some platforms do not protect user data properly.
- Unauthorized access to personal data is a major concern.
- Developers building simple goal management apps often struggle with authentication and error handling implementation.

As a result, users feel insecure about storing personal goals online, and developers face challenges implementing secure backend systems.

Solution

A **Node.js and Express-based Goal Management Backend API** that:

- Provides secure **User Registration and Login** functionality.
- Uses **JWT authentication** to protect API routes.
- Implements full **CRUD operations** for goal management.
- Ensures centralized **error handling middleware** for consistent responses.
- Stores data securely in **MongoDB** using structured models.
- Follows a modular architecture (Routes → Controllers → Models → Middleware) for scalability.